

MILP Formulation for Agroforestry Biomass Supply Chain (v10)

Multi-Cycle Harvesting with Age Lag Constraints

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1 1. Index Sets

$i \in \mathcal{I} = \{1, \dots, n_s\}$	Agroforestry sites (1)
$j \in \mathcal{J} = \{1, \dots, n_j\}$	Storage/processing facilities (2)
$k \in \mathcal{K} = \{1, \dots, n_k\}$	Consumer sites (3)
$t \in \mathcal{T} = \{1, \dots, T_{\max}\}$	Time periods (years) (4)
$t \in \mathcal{T}^+ = \{0, \dots, T_{\max} + 1\}$	Time periods (years) (5)
$t \in \mathcal{T}^{\langle \neg \nabla \sqsubseteq \rangle} = \{A^{\min} + 1, \dots, T_{\max}\}$	Harvesting periods (years) (6)
$p \in \mathcal{P} = \{1, 2, 3\}$	Product types (chemical, pulp, energy) (7)
$(s, t) \in \mathcal{S} = \{(s, t) s, t \in 0, \dots, (T_{\max} + 1) \wedge (t - s) \in \{A^{\min}, \dots, A^{\max}\}\}$	Consecutive harvesting (8)

2 Example set $\mathcal{S}$

let $A^{\min} = 1$, $A^{\max} = 5$ and $T_{\max} = 10$, then $\mathcal{S}$ includes pairs like:

Subset	quantifier	Description
$(0, t)$	$t = 1, \dots, 9$	establishment arcs
$(t, 11)$	$t = 1, \dots, 10$	closing arcs
$(1, t)$	$t = 2, \dots, 6$	cultivation arcs of period 1 (harvest in t)
$(2, t)$	$t = 3, \dots, 7$	cultivation arcs of period 2 (harvest in t)
$(3, t)$	$t = 4, \dots, 8$	cultivation arcs of period 3 (harvest in t)
$(4, t)$	$t = 5, \dots, 9$	cultivation arcs of period 4 (harvest in t)
$(5, t)$	$t = 6, \dots, 10$	cultivation arcs of period 5 (harvest in t)
$(6, t)$	$t = 7, \dots, 10$	cultivation arcs of period 6 (harvest in t)
$(7, t)$	$t = 8, \dots, 10$	cultivation arcs of period 7 (harvest in t)
$(8, t)$	$t = 9, \dots, 10$	cultivation arcs of period 8 (harvest in t)
$(9, 10)$	—	cultivation arc of period 9 (harvest in 10)

If establishment occurs in period 3 and harvests in periods 6 and 10, the following variables

would be active:

- $z_{i,0,3} = 1$ (establishment),
- $z_{i,3,6} = 1$ (harvest in 6 with establishment in 3),
- $z_{i,6,10} = 1$ (harvest in 10 with predecessor in 6),
- $z_{i,10,11} = 1$ (last harvest in 10), and
- $z_{ist} = 0$ for all $(s, t) \in \mathcal{S}$

Constraints C1 and C2 read like this

- C1: $\sum_{t=1}^{10} z_{i,0,t} \leq 1$ (at most one establishment) $\rightarrow$ as $z_{i,0,3} = 1$ it follows that $z_{i,0,t} = 0$ for all $t \neq 3$.
- C2_1: $z_{i,0,1} = \sum_{u \in \{2, \dots, 6, 11\}} z_{i,1,u}$ (if there is establishment in 1, there must be an harvest in u or closing) $\rightarrow$ as $z_{i,0,1} = 0$, $\sum_{u \in \{2, \dots, 6, 11\}} z_{i,1,u} = 0$.
- C2_2: $z_{i,0,2} + z_{i,1,2} = \sum_{u \in \{3, \dots, 7, 11\}} z_{i,2,u}$ (if there is establishment or harvest in 2, there must be an harvest in u or closing) $\rightarrow$ as $z_{i,0,2} = z_{i,1,2} = 0$, $\sum_{u \in \{3, \dots, 7, 11\}} z_{i,3,u} = 0$
- C2_3: $z_{i,0,3} + z_{i,1,3} + z_{i,2,3} = \sum_{u \in \{4, \dots, 8, 11\}} z_{i,3,u}$ (if there is establishment or harvest in 3, there must be an harvest in u or closing) $\rightarrow$ as $z_{i,0,3} = 1$ and $z_{i,1,3} = z_{i,2,3} = 0$, $\sum_{u \in \{4, \dots, 8, 11\}} z_{i,3,u} = 1 \rightarrow$ as $z_{i,3,6} = 1$, it follows that $z_{i,3,u} = 0$ for all $u \neq 6$.
- C2_4: $z_{i,0,4} + z_{i,1,4} + z_{i,2,4} + z_{i,3,4} = \sum_{u \in \{5, \dots, 9, 11\}} z_{i,4,u}$ (if there is establishment or harvest in 4, there must be an harvest in u or closing) $\rightarrow$ as $z_{i,0,4} + z_{i,1,4} + z_{i,2,4} + z_{i,3,4} = 0$, it follows that $\sum_{u \in \{5, \dots, 9, 11\}} z_{i,4,u} = 0$.
- and so on

3 2. Decision Variables

3.1 2.1 Binary Variables

$z_{ist} \in \{0, 1\}$ Indicator if at site i in period st , s are consecutive harvests with $(s, t) \in \mathcal{S}$ (9)

3.2 2.2 Continuous Variables

$X_{ijpt} \geq 0$ Flow of product $p \in \mathcal{P}$ from site $i \in \mathcal{I} \rightarrow$ storage $j \in \mathcal{J}$ in period $t \in \mathcal{T}^{(-\nabla \sqsubseteq)}$ [tonnes] (10)

$X_{jkpp't} \geq 0$ Conversion flow of product $p \in \mathcal{P}$ to product $p' \in \mathcal{P} : p' \geq p$ from storage $j \rightarrow$ consumer k in period t (11)

$Y_{ipt} \geq 0$ Maximum harvest quantity of product p at site i in period $t \in \mathcal{T}^{(-\nabla \sqsubseteq)}$ [tonnes] (12)

$S_{jpt} \geq 0$ Inventory at storage j in period $t \in \mathcal{T}$ [tonnes] (13)

4 3. Parameters

Parameter	Unit	Description	Typical Value
A^{min}	yr	minimum age	3–5 years
A^{max}	yr	maximum age	8–10 years
$AREA_i$	ha	Site area	8–20 ha
c_i^{est}	€/ha	Establishment cost in € per ha	10 €/ha
c_i^{opp}	€/ha	Opportunity cost per ha	5 – 15 €/ha
c_i^{harv}	€/ha	Harvest and refitting cost per ha	5 – 15 €/ha
η_{pa}	t/ha	Base yield for product p , age a	4–60 t/ha
$R_{k,p}$	€/t	Revenue for product p by consumer k	80–250 €
$c^{tr-raw/pre}$	€/t/km	Transport cost rates per ton and kilometer for raw and pre-treated biomass	0.10/0.15 €/km-ton
d_{ij}, d_{jk}	km	Distances	30–200 km
CAP_j^{stor}	t	Storage capacity	300–500 t
CAP_j^{proc}	t/yr	Processing capacity	80–150 t/yr
D_{kp}^{max}	t/yr	Demand at consumer k	Varies

5 4. Objective Function

$$\max Z = \sum_{k,p,t} R_{k,p} \cdot \sum_{j \in \mathcal{I}, p' \in \mathcal{P}: p' \leq p} X_{jkp't} \quad \text{Revenue} \quad (14)$$

$$- \sum_i C_i^{est} \cdot \text{AREA}_i \cdot \sum_{t \in T} z_{i0t} \quad \text{Establishment} \quad (15)$$

$$- \sum_i C_i^{opp} \cdot \text{AREA}_i \cdot \sum_{t \in T} (T^{max} - t) \cdot z_{i0t} \quad \text{Opportunity} \quad (16)$$

$$- \sum_{i, (s,t) \in \mathcal{S} | s > 0} C_i^{harv} \cdot \text{AREA}_i \cdot z_{ist} \quad \text{Harvest} \quad (17)$$

$$- c^{tr-raw} \cdot \sum_{i,j,p,t} d_{ij} \cdot X_{ijpt} \quad \text{Transport (site} \rightarrow \text{hub)} \quad (18)$$

$$- c^{tr-pre} \cdot \sum_{j,k,p,t} d_{jk} \cdot X_{jkpt} \quad \text{Transport (hub} \rightarrow \text{consumer)} \quad (19)$$

$$- \sum_{j,p,t} c_j^{stor} \cdot S_{jpt} \quad \text{Storage} \quad (20)$$

6 5. Constraints

6.1 Constraint Set 1: AFS establishment

$$\sum_{t \in \mathcal{T}^+} z_{i0t} \leq 1 \quad \forall i \in \mathcal{I} \quad (C1)$$

Interpretation: If AFS is established in t , $z_{i,0,t} = 1$.

6.2 Constraint Set 2: Path connectivity

$$\sum_{(s,t) \in \mathcal{S}} z_{ist} = \sum_{(t,u) \in \mathcal{S}} z_{itu} \quad \forall i \in \mathcal{I}, t \in \mathcal{T} \quad (C2)$$

Interpretation: If there is a harvest in t , there must be an predecessor in $s < t$ and successor in $u > t$.

6.3 Constraint Set 3: Biomass yield calculation

$$Y_{ipt} = \sum_{s=1}^{t-1} \eta_{p(t-s)} \cdot AREA_i \cdot z_{ist} \quad \forall i \in \mathcal{I}, p \in \mathcal{P}, t \in \mathcal{T}^{(+\nabla)} \quad (C3)$$

(21)

Interpretation: Biomass yield is restricted by age of plantage

6.4 Constraint Set 4: Yield Shipping Linkage

$$\sum_{j \in \mathcal{J}} X_{ijpt} \leq Y_{ipt} \quad \forall i \in \mathcal{I}, p \in \mathcal{P}, t \in \mathcal{T}^{(+\nabla)} \quad (C4)$$

Interpretation: Biomass shipped from site i (product p , period t) is limited by available harvest.

6.5 Constraint Set 5: Inventory Balance

$$S_{jpt} = S_{jp(t-1)} + \sum_i X_{ijpt} - \sum_{k, p' \geq p} X_{jkpp't} \quad \forall j \in \mathcal{J}, p \in \mathcal{P}, t \in \mathcal{T}^{(+\nabla)} \quad (C5)$$

with $S_{j,p,0} = 0$.

Interpretation: Flow conservation at storage hubs.

6.6 Constraint Set 6: Storage Capacity

$$\sum_{p \in \mathcal{P}} S_{jpt} \leq CAP_j^{stor} \quad \forall j \in \mathcal{J}, t \in \mathcal{T}^{(+\nabla)} \quad (C6)$$

Interpretation: Total inventory limited by physical storage space.

6.7 Constraint Set 7: Processing Capacity

$$\sum_{i \in \mathcal{I}, p \in \mathcal{P}} X_{ijpt} \leq CAP_j^{proc} \quad \forall j \in \mathcal{J}, t \in \mathcal{T}^{(+\nabla)} \quad (C7)$$

Interpretation: Incoming material throughput limited by processing equipment.

6.8 Constraint Set 8: Demand Satisfaction with Product Cascade

$$\sum_{j \in \mathcal{J}, p' \in \mathcal{P}: p' \leq p} X_{jkp't} \leq D_{kpt}^{max} \quad \forall k \in \mathcal{K}, p \in \mathcal{P}, t \in \mathcal{T}^{(\neg \nabla \Xi)} \quad (\text{C8})$$

Interpretation: Higher-quality products (lower p) can substitute for lower-quality demand.

Cascade hierarchy: - Product 1 (chemical) → Satisfies any demand - Product 2 (pulp) → Satisfies product 2 or 3 demand - Product 3 (energy) → Only satisfies product 3 demand
